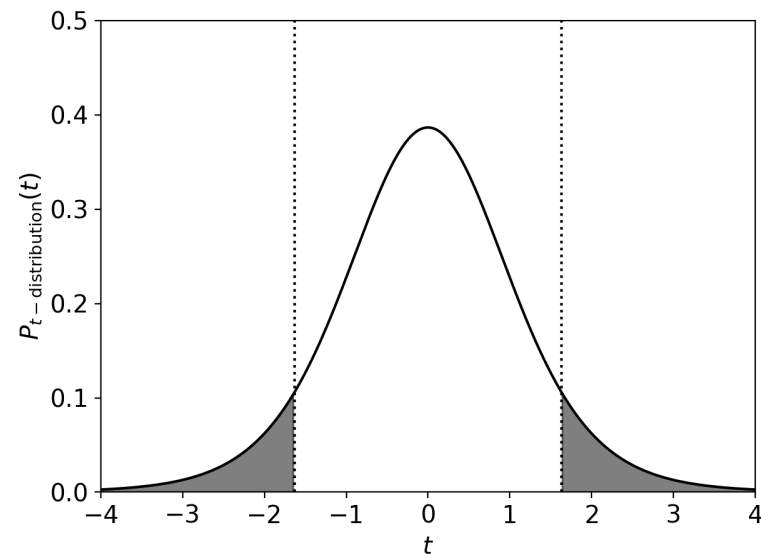
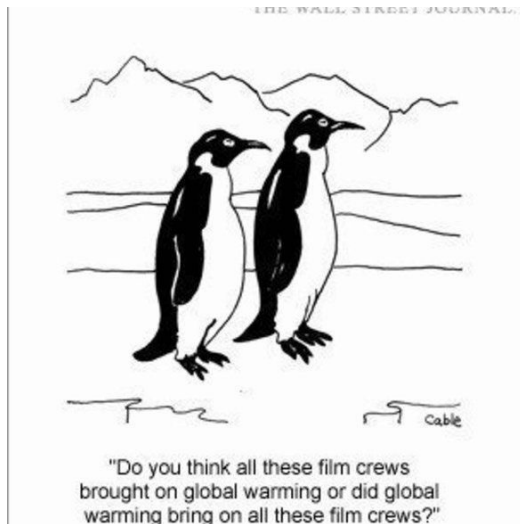
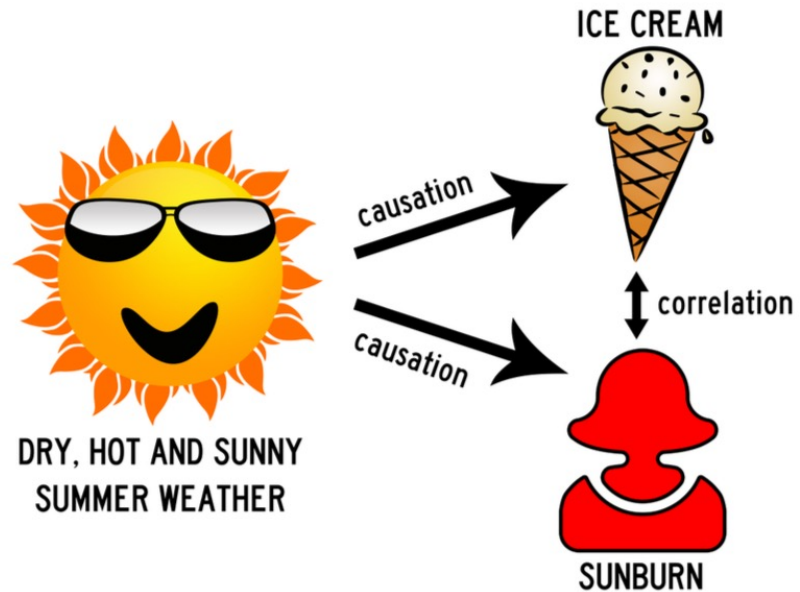
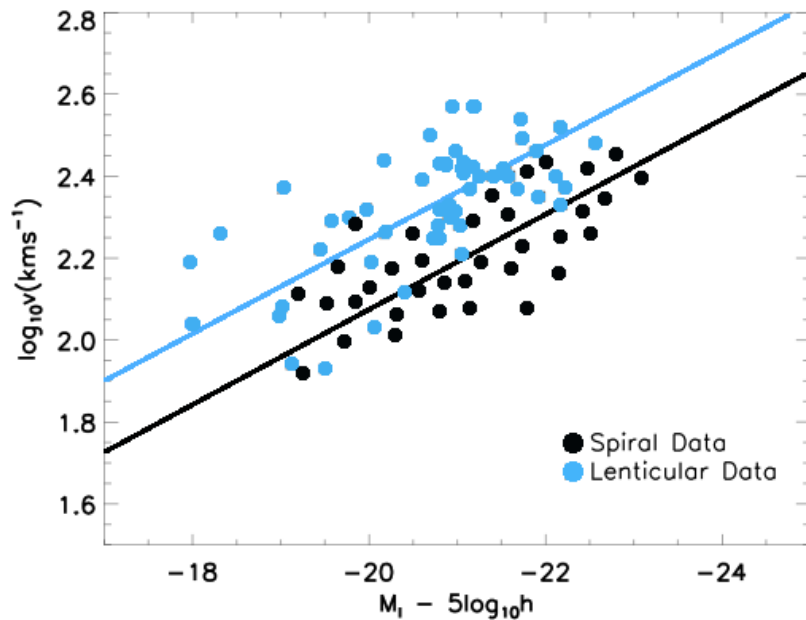
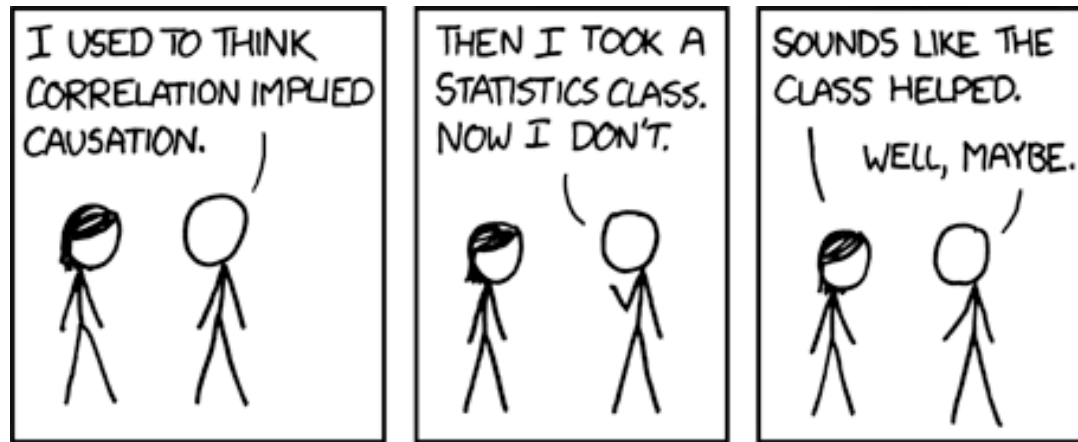


Class 2: Correlation Testing

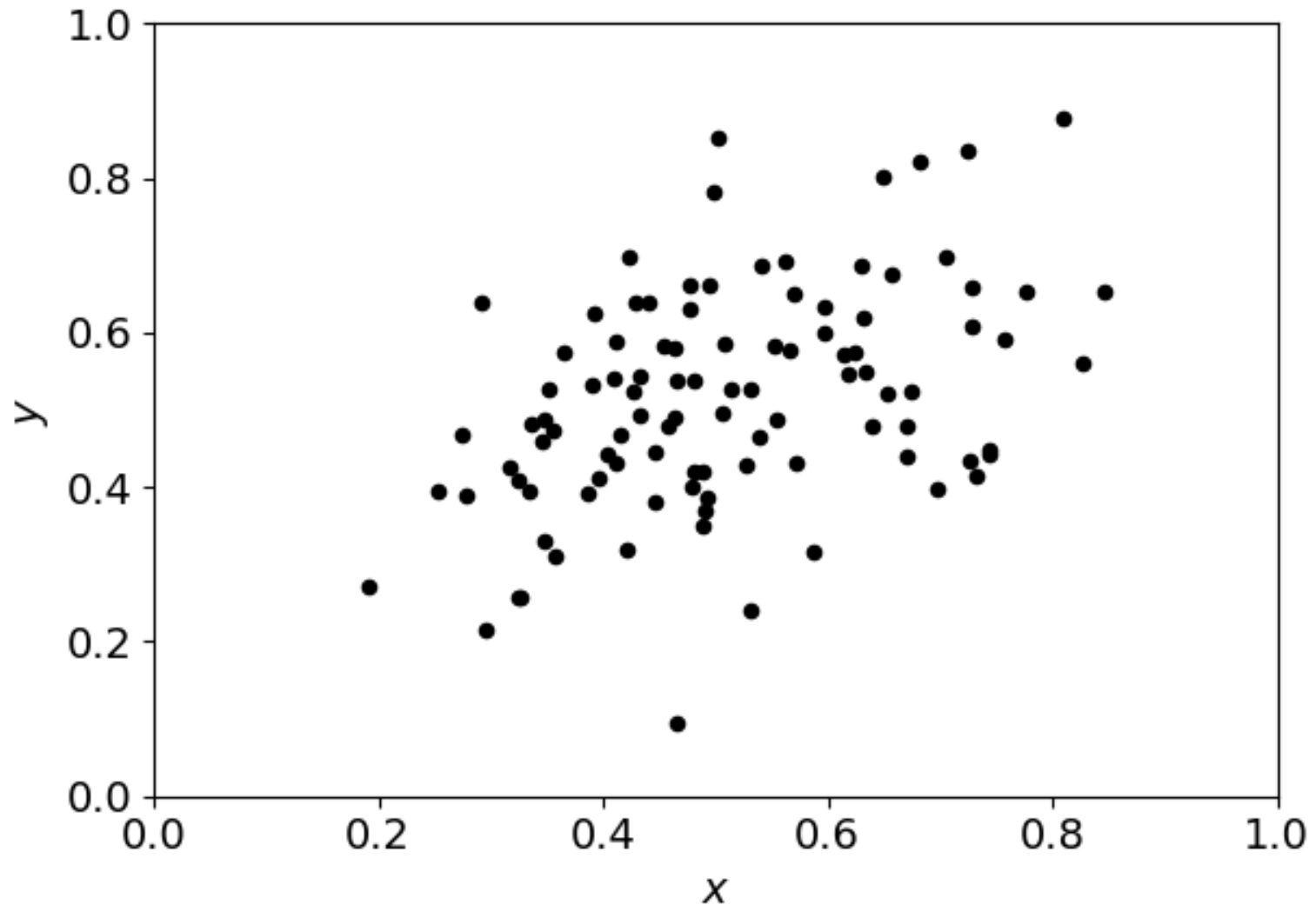


Quick intro to statistics course



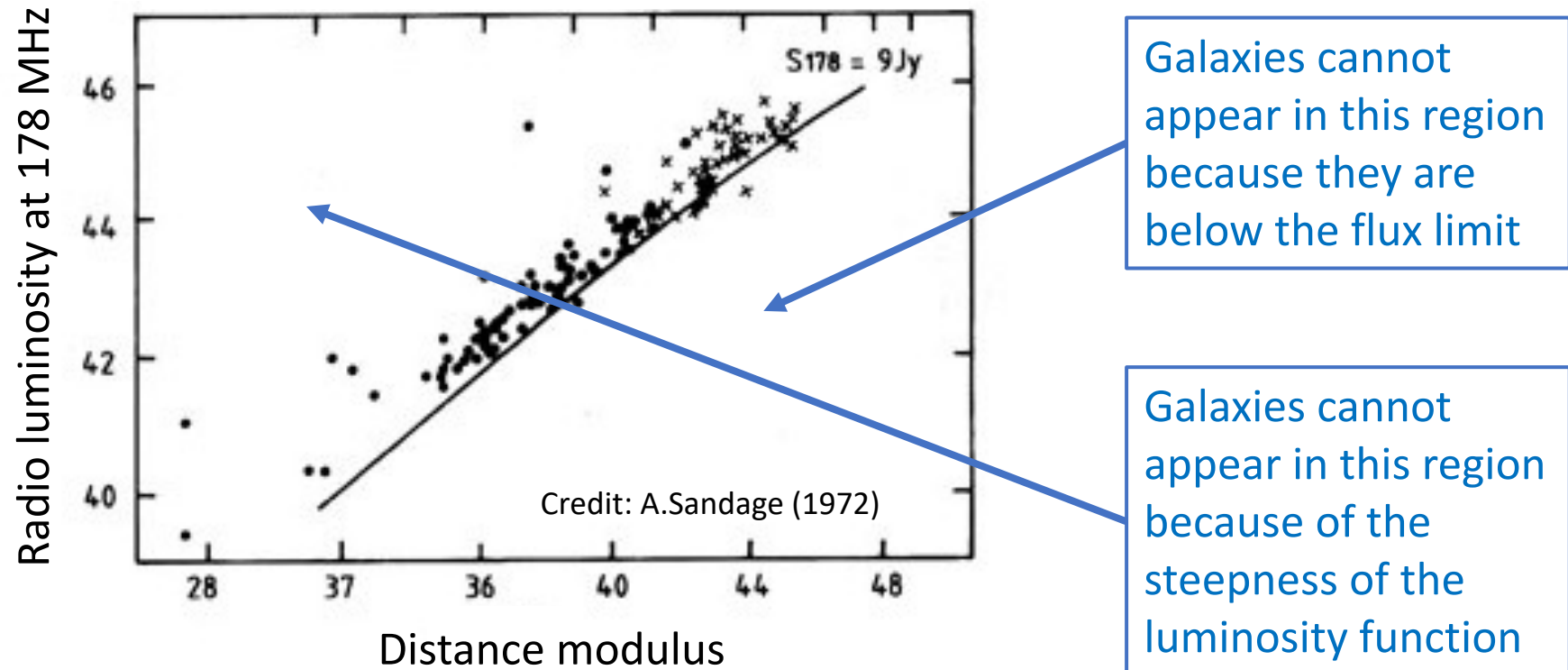
- Week 1: Probability and statistics
- Week 2: Correlation testing
- Week 3: Model fitting
- Week 4: Regression
- Week 5: Error estimates
- Week 6: Bayesian methods

Searching for correlations!



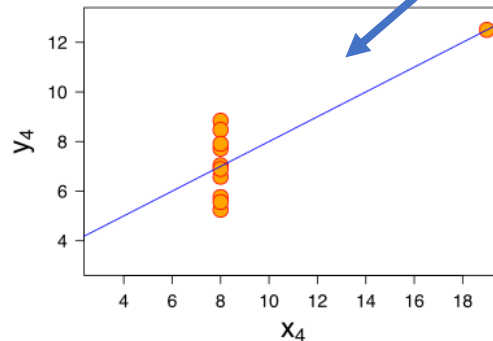
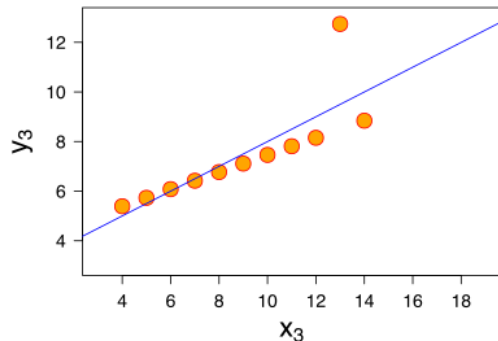
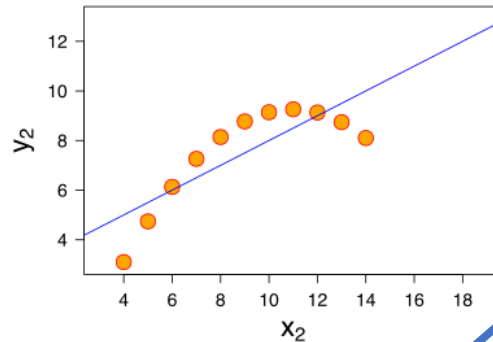
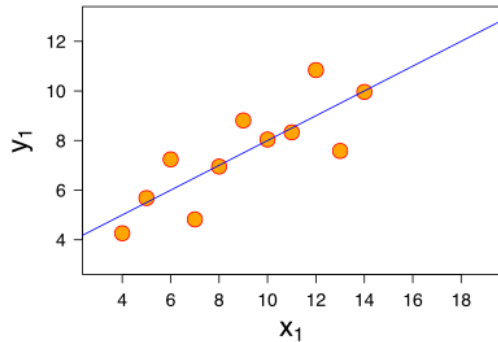
Pitfalls when searching for correlations

- **Selection effects** can easily lead to spurious correlations
- *Here is a perfect luminosity-redshift correlation for radio galaxies in the 3CR survey:*



Pitfalls when searching for correlations

- Correlations can be driven by a **small number of outliers**
- *The following four (x, y) datasets all have the same mean, variance, correlation coefficient and regression line:*



The correlation here is completely driven by a single outlier

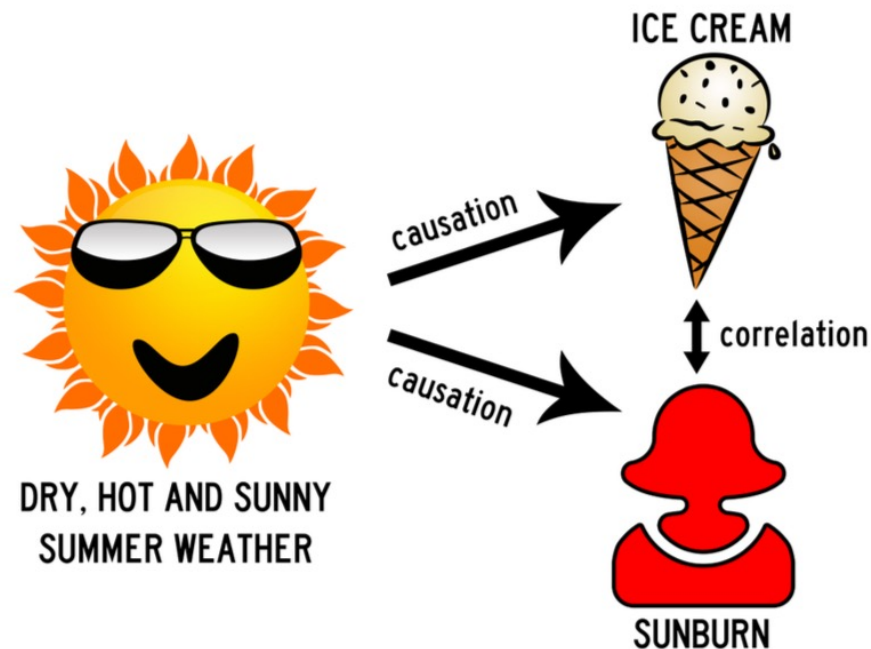
“Rule of thumb”: if a correlation goes away after you cover part of the dataset with your thumb, it probably isn’t real!



Pitfalls when searching for correlations

- **Correlation is not the same as causation**
- The correlation of two variables does not necessarily imply a causal/direct connection. They might both be driven by a **“third variable”**.

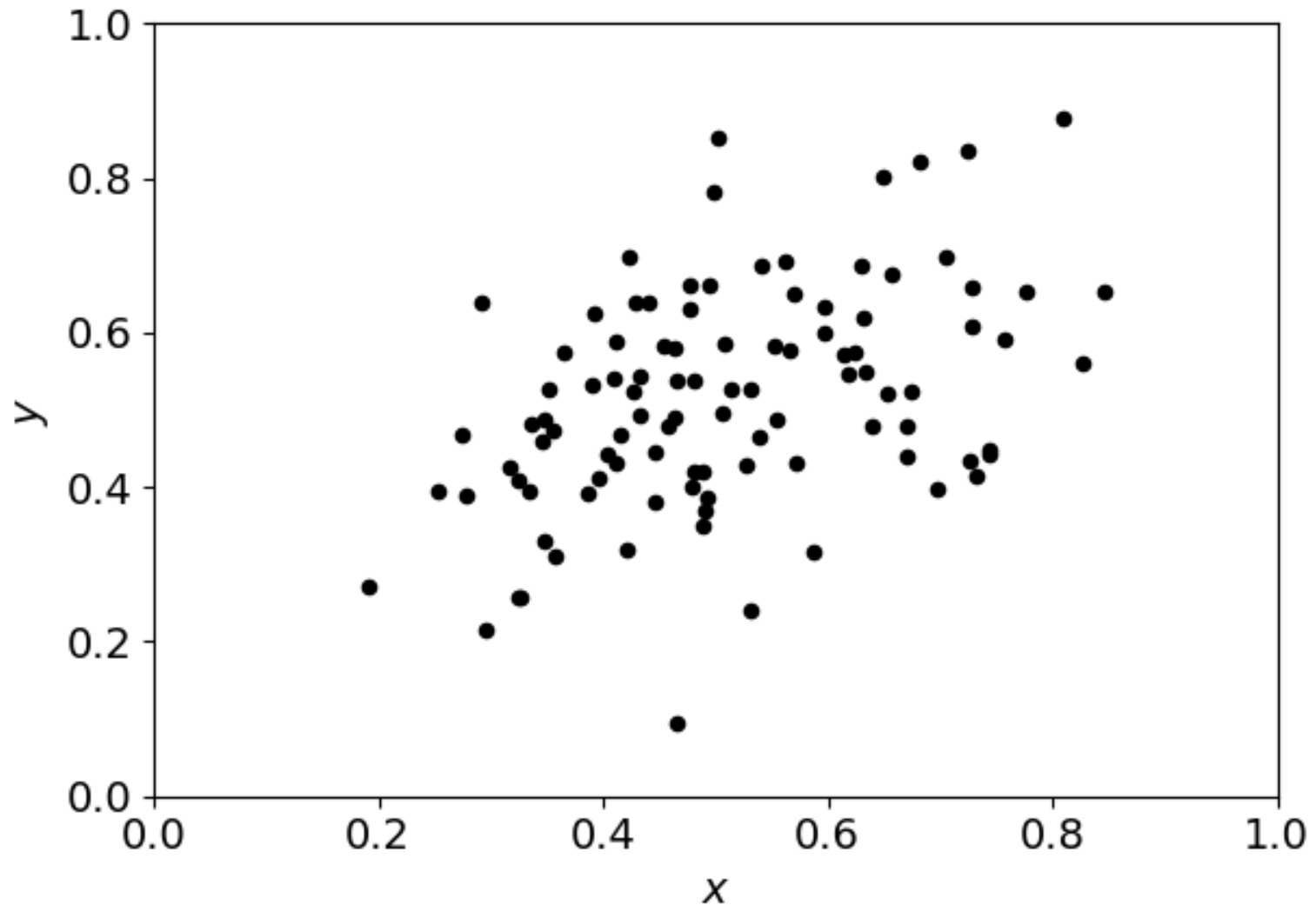
*Eating ice cream
causes sunburn??*



Procrastinate by checking a few more examples at <https://www.tylervigen.com/spurious-correlations>

Credit: <https://towardsdatascience.com/correlation-is-not-causation-ae05d03c1f53>

Searching for correlations!



Significance of correlation

- We can estimate the correlation coefficient of data samples (x_i, y_i) using the **Pearson product-moment** formula:

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}} = \frac{\sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}}{(N - 1) \sqrt{\text{Var}(x) \text{Var}(y)}}$$

(Or rather, we get python to evaluate this formula!!)

```
from scipy import stats
r,p = stats.pearsonr(x,y) # evaluate Pearson correlation coefficient r and p-value for non-correlation
print('Correlation coefficient r =',r)
print('Error in r =',np.sqrt((1.-r**2)/float(n-2)))
print('p-value for non-correlation =',p)
```

```
Correlation coefficient r = 0.548341521611091
Error in r = 0.08447446031603906
p-value for non-correlation = 3.513624460312244e-09
```

Significance of correlation

- We can estimate the correlation coefficient of data samples (x_i, y_i) using the **Pearson product-moment** formula:

$$r = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}} = \frac{\sum_{i=1}^N x_i y_i - N \bar{x} \bar{y}}{(N - 1) \sqrt{\text{Var}(x) \text{Var}(y)}}$$

- When correlation-testing, it is **not** sufficient to just measure r . We also need to check the **significance of the correlation**
- *Correlations can arise by random chance!*
- We can test the significance of the correlation by carrying out a **hypothesis test**

Significance of correlation

- *How do we apply a hypothesis test to correlations?*
- **Null hypothesis:** there is no correlation between the variables
- **Test statistic:**
$$t = r \sqrt{\frac{N - 2}{1 - r^2}}$$

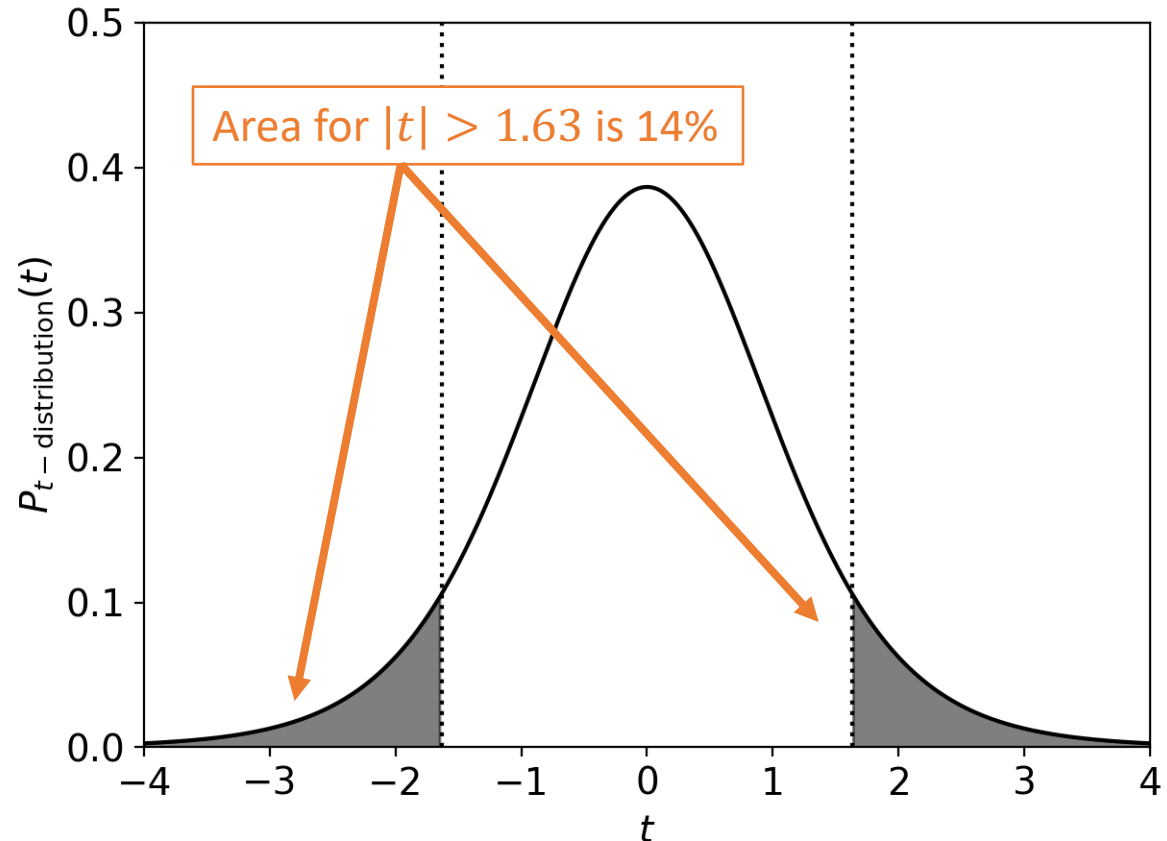
r = correlation coefficient
 N = number of data points
(`scipy.stats` is doing all this for us...)
- **Distribution followed by the statistic:** the Student's t probability distribution with number of degrees of freedom $\nu = N - 2$
- **Probability of rejecting the hypothesis:** the area under the tails at higher values of $|t|$ than we have measured

Worked example: correlation test

- *Example: we measure $r = 0.5$ for $N = 10$ points. Is this correlation significant?*

If we carried out this experiment many times, this is the distribution that would result!

- We find $t = 1.63$, $\nu = 8$
- The probability of finding $|t| > 1.63$ is **14%**
- This is **not** sufficiently small to reject the hypothesis of no correlation: **this correlation is not significant**
- [we would typically reject with (e.g.) 95, 99% confidence]



Activity: Expansion #1

- In this Activity we will check **who discovered the expansion of the Universe!** See Hubble and Lemaitre's distance-velocity datasets. For the two datasets, determine the **Pearson correlation coefficient**, its **error** and **statistical significance**

```
In [5]: # Read in datasets
data = np.loadtxt('../datasets/lemaitre.dat')
dl,vl = data[:,0],data[:,1]
data = np.loadtxt('../datasets/hubble.dat')
dh,vh = data[:,0],data[:,1]
nl,nh = len(dl),len(dh)

# Pearson correlation analysis of Lemaitre dataset
rl,pl = stats.pearsonr(dl,vl)
rlerr = np.sqrt((1.-(rl**2))/float(nl-2))
print('Lemaitre dataset:')
print('r =',rl)
print('error in r =',rlerr)
print('p-value =',pl)

# Pearson correlation analysis of Hubble dataset
rh,ph = stats.pearsonr(dh,vh)
rherr = np.sqrt((1.-(rh**2))/float(nh-2))
print('Hubble dataset:')
print('r =',rh)
print('error in r =',rherr)
print('p-value =',ph)
```

Lemaitre dataset:

- $r = 0.38$
- Error in $r = 0.14$
- $p\text{-value} = 0.013$

Hubble dataset:

- $r = 0.79$
- Error in $r = 0.13$
- $p\text{-value} = 4.5 \times 10^{-6}$

Hypothesis tests and p -values

- The probability of rejecting a hypothesis is often known as a “ p -value”
- It corresponds to the “**significance**” of a result
- *Let's talk about exactly what this value means, since this can be pretty confusing*

<u>P-VALUE</u>	<u>INTERPRETATION</u>
0.001	HIGHLY SIGNIFICANT
0.01	
0.02	
0.03	
0.04	SIGNIFICANT
0.049	
0.050	OH CRAP. REDO CALCULATIONS.
0.051	ON THE EDGE OF SIGNIFICANCE
0.06	
0.07	HIGHLY SUGGESTIVE, SIGNIFICANT AT THE $P < 0.10$ LEVEL
0.08	
0.09	
0.099	HEY, LOOK AT THIS INTERESTING SUBGROUP ANALYSIS
≥ 0.1	

Hypothesis tests and p -values

- Suppose a (no-) correlation significance yields $p = 0.01$
- This means: **there is a 1% chance of obtaining a set of measurements at least this correlated, if the underlying data is uncorrelated.** It does not mean:
 - “the probability that the points are uncorrelated is 1%”
 - “the probability that the points are correlated is 99%”
 - “if we claim a correlation, there is a 1% chance that we would be mistaken”
- **Frequentist statistics cannot assess the probability that the model itself is correct** (see – Bayesian statistics)

Non-parametric correlation tests

- If we do not want to assume that (x, y) are drawn from a bivariate Gaussian, we can use a **non-parametric correlation test**
- Let (X_i, Y_i) be the rank of (x_i, y_i) in the overall order, such that $1 \leq X_i \leq N$ and $1 \leq Y_i \leq N$
- Compute the **Spearman rank correlation coefficient**

$$r_s = 1 - 6 \frac{\sum_{i=1}^N (X_i - Y_i)^2}{N^3 - N}$$

- Convert the correlation coefficient into a **probability**, using the Student's t distribution as before, with number of degrees of freedom $\nu = N - 2$

Bayesian correlation methods

- To determine the significance of our correlation, we have been asking, “**what is the probability of measuring a particular value of r if there is no correlation?**”

Mathematically,

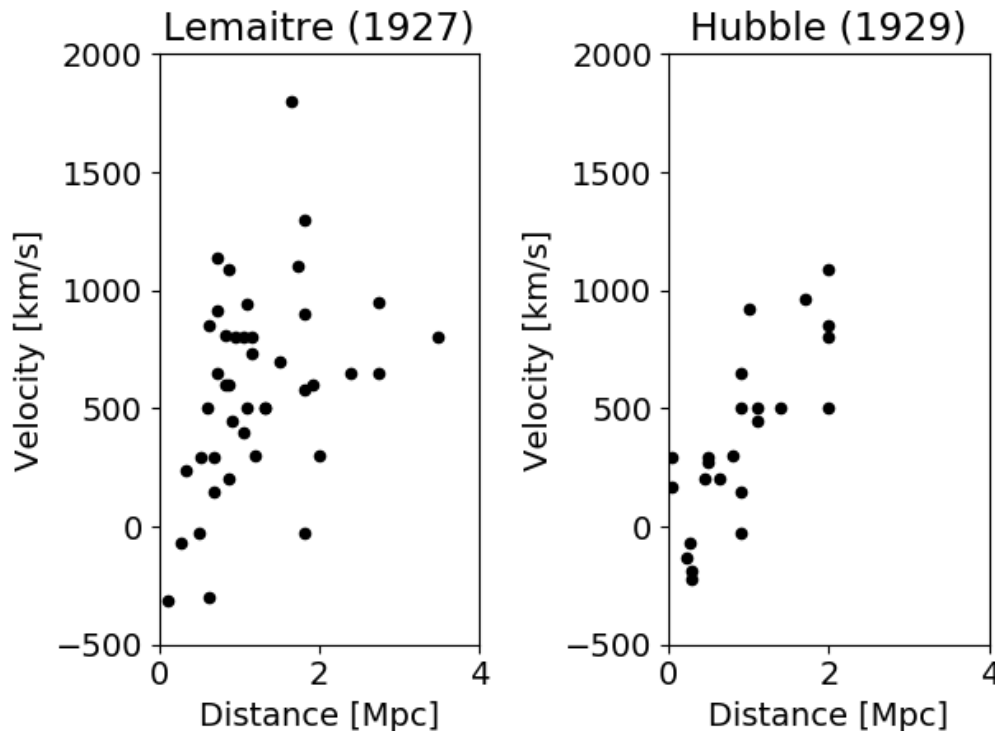
$$P(r|\rho = 0)$$

- Using Bayesian statistics we can ask the opposite question: “**what is the posterior probability distribution for the correlation coefficient ρ given the measured value of r ?**”

$$P(\rho|r) \propto \frac{(1 - \rho^2)^{\frac{N-1}{2}}}{(1 - \rho r)^{N-\frac{3}{2}}} \left(1 + \frac{1}{N - \frac{1}{2}} \frac{1 + \rho r}{8} + \dots \right)$$

Activity: Expansion #2

- Returning to Hubble and Lemaitre's distance-velocity datasets, now determine the **Spearman rank correlation coefficient**, its **statistical significance**, and the full **probability distribution of $P(\rho|r)$** using the Bayesian formula.



Lemaitre dataset:

- $r = 0.42$
- $p\text{-value} = 0.006$

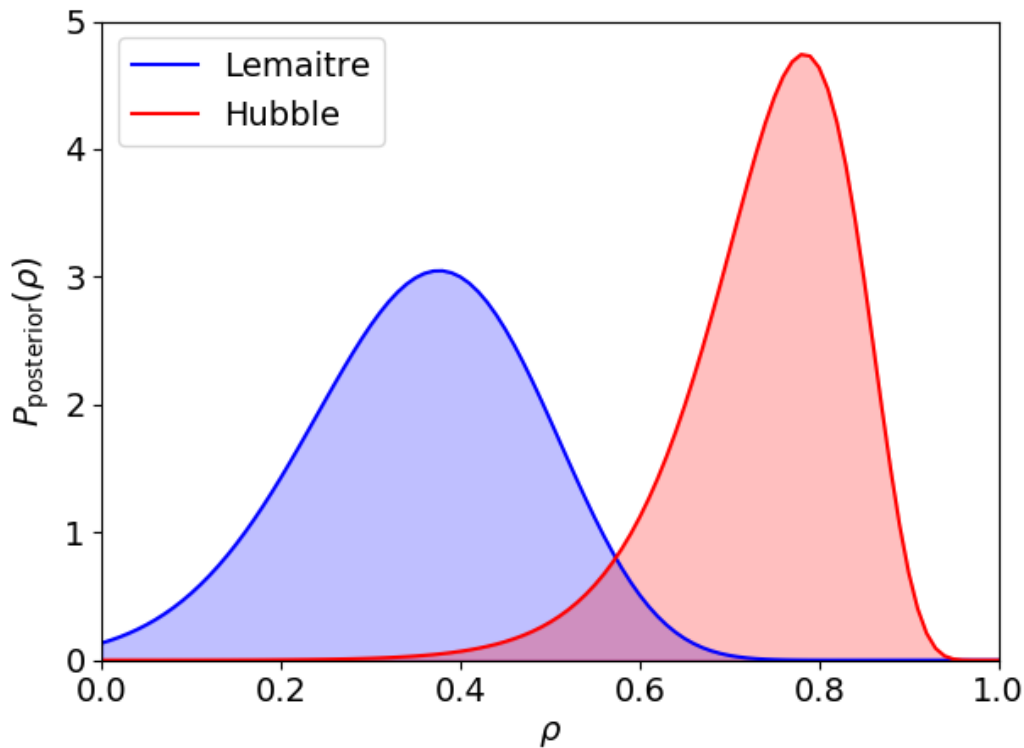
Hubble dataset:

- $r = 0.80$
- $p\text{-value} = 3.4 \times 10^{-6}$

Excellent example of why we need to specify test **before** experiment

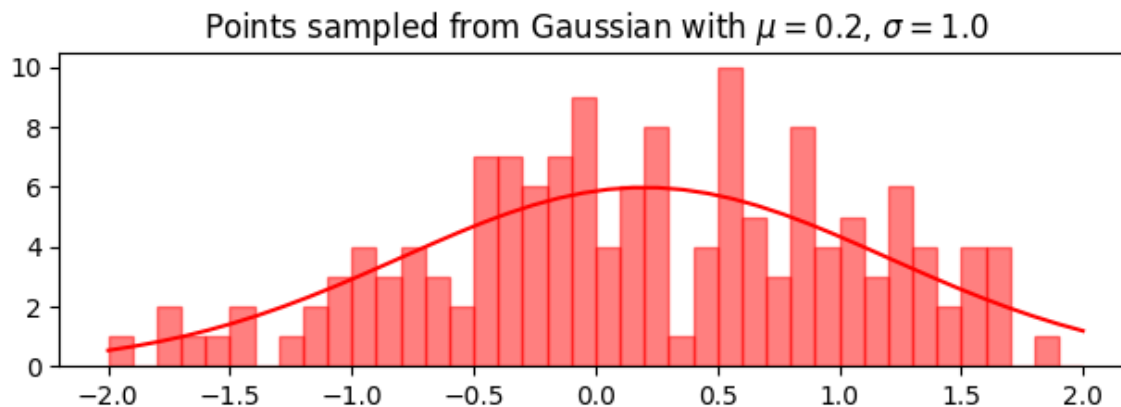
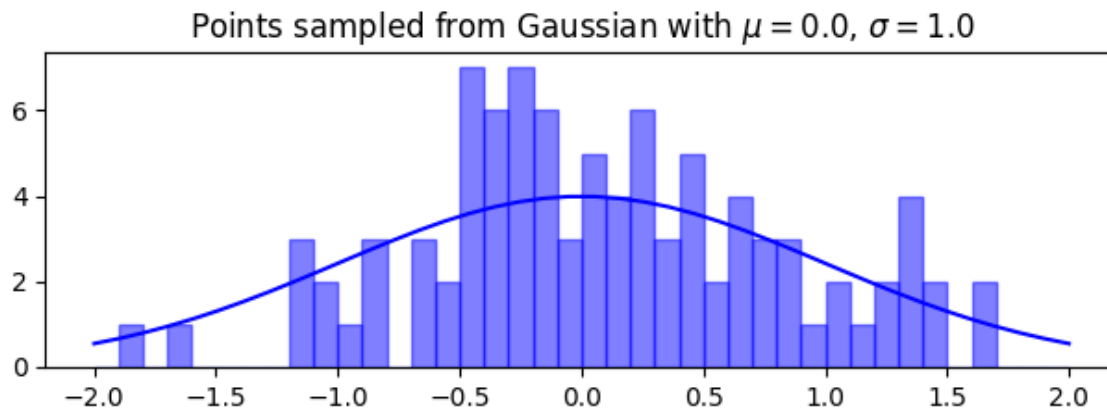
Activity: Expansion #2

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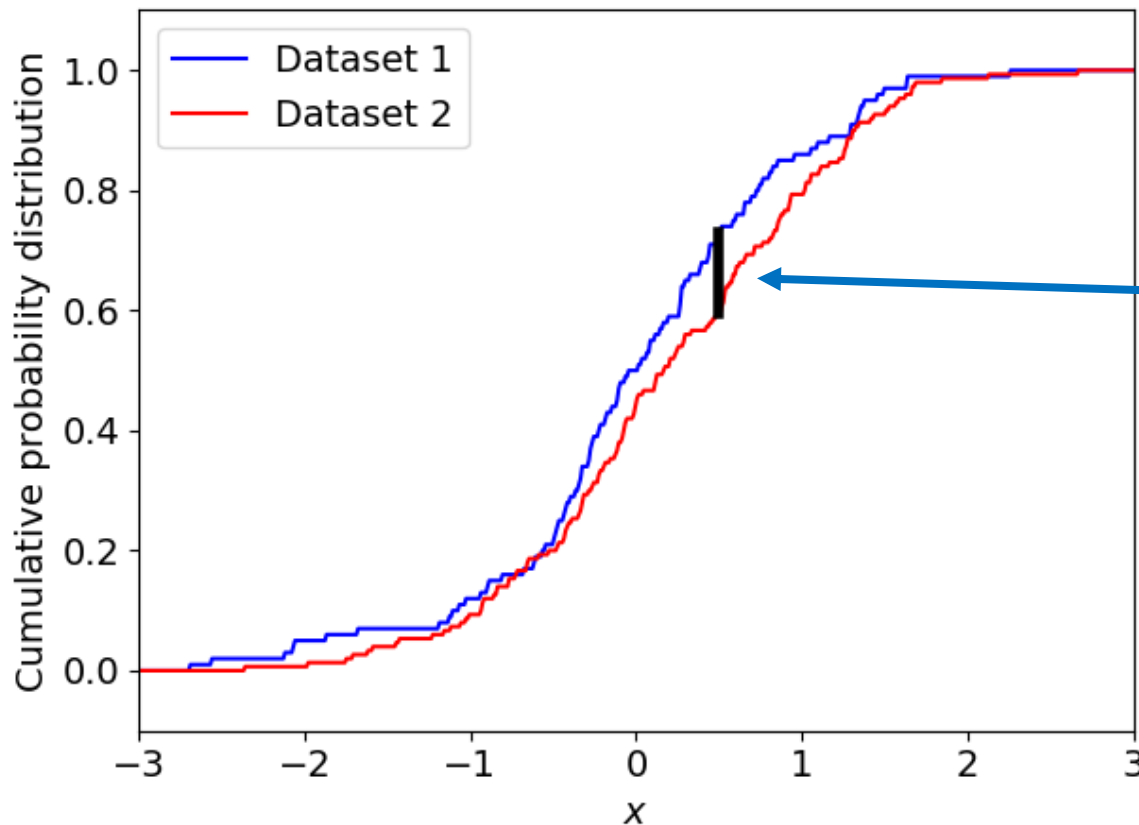
Worked example: K-S test

- To test whether two full distributions are consistent (that is, drawn from the same parent distribution) we can use the **Kolmogorov-Smirnov (K-S) test**



Worked example: K-S test

- Cumulative probability distribution (going from left to right of the histograms):



- The null hypothesis is that these datasets are drawn from the **same parent distribution**
- This is the **maximum deviation**, $d = 0.14$
- The **probability** of rejecting the null hypothesis is $p = 0.196$
- *We confirm the hypothesis!*

Activity: K-S test

- The provided datasets list estimated masses for neutron stars which **are** in double neutron star binaries and are **not** in double neutron star binaries

- Use the t -test to determine whether *there is any significant difference in the means of the*

My answers:

p -value for same means =

- Use
deter
mas
cons
- ```
datfile1 = '../datasets/ns_masses1.dat'
datfile2 = '../datasets/ns_masses2.dat'
x1 = np.loadtxt(datfile1)
x2 = np.loadtxt(datfile2)
result = stats.ttest_ind(x1,x2)
print('p-value for same means =',result.pvalue)
result = stats.ks_2samp(x1,x2)
print('p-value for same parent distribution =',result.pvalue)
```

# Next week: Model-fitting

